

```
8 root :0 :0 Wed Oct 11 12:56
9 (unknown :0 :0 Wed Oct 11 12:55
10 reboot system 3.10.0-123.el7.x Thu Oct 12 12:37
```

The data shows that the data frame is not in a uniform table format and fields of records are separated by a tab character. This requires some amount of filtering of data in the table to extract relevant rows for further processing. Since I will be demonstrating the distribution of IP addresses within a system, only the IP address and other related fields are kept for histogram plotting.

To have a statistical evaluation of this data, it is worth removing all the irrelevant fields from the data frame:

```
drops = c(colnames(rfi)[2], colnames(rfi)[3], colnames(rfi)[5],
colnames(rfi)[6], colnames(rfi)[7], colnames(rfi)[8],
colnames(rfi)[9], colnames(rfi)[10], colnames(rfi)[11])
d = rfi[, !(names(rfi) %in% drops)]
```

Then, for simplicity, extract all IP addresses attached to the user 'ccadmin' which start with '172'.

```
u = d[like(d$user, 'ccadmin') & like(d$ipaddress, '172'),]
```

Now the data is ready for analysis. The R *summary* command will show the count of elements of each field, whereas the *count* command will show the frequency distribution of the IP address as shown below:

```
> summary(u)
 user      ipaddress
ccadmin  :34  172.16.6.252:21
```

```
: 0 172.16.7.155: 6
(unknown : 0 172.16.5.230: 3
backup_u : 0 172.16.11.95: 2
reboot : 0 172.16.4.66 : 1
root : 0 172.16.5.132: 1
(Other) : 0 (Other) : 0
```

```
and
> count(u)
 user      ipaddress      freq
1 ccadmin 172.16.11.95      2
2 ccadmin 172.16.4.66      1
3 ccadmin 172.16.5.132      1
4 ccadmin 172.16.5.230      3
5 ccadmin 172.16.6.252     21
6 ccadmin 172.16.7.155      6
```

For better visualisation, this frequency distribution of the IP address can be depicted using a histogram, as follows:

```
qplot(u$ipaddress, main='IP histogram', xlab='BioMass of
Leaves', ylab='Frequency')
```



By: Dipankar Ray

The author is a member of IEEE and IET, and has more than 20 years of experience in open source versions of UNIX operating systems and Sun Solaris. He is presently working on data analysis and machine learning using a neural network as well as on different statistical tools. He has also jointly authored a textbook called 'MATLAB for Engineering and Science'. He can be reached at dipankarray@ieee.org.

Read more stories on Components in
www.electronicb2b.com

TOP COMPONENTS STORIES

- The latest in power converters
- India's leading component distributors
- Growth of Indian electronics components industry
- The latest launches of components for LEDs
- The latest launches of components for electronics

**ELECTRONICS
B2B**
INDUSTRY IS AT A

CLICK OF A BUTTON

Log on to www.electronicb2b.com and be in touch with the Electronics B2B Fraternity 24x7